

BRAINSTORMING

Learning Objectives

After reading this brief, you will be able to...

- I. Understand the rules of brainstorming
- II. Know how to plan and lead a brainstorming session
- III. Capture the output of a brainstorm to support next steps in the biodesign process

Getting Started

Finally, it's time to think about solutions! You've tirelessly worked to get your need statement just right, and you've validated your need criteria with the most important stakeholders. Now all of that hard work is about to pay off.

The need statement and criteria are the jumping-off point for the next major phase of the Biodesign process: INVENT. And the first thing you do during INVENT is concept generation – or ideating new ideas.

A well-formed need statement is designed to trigger ideation, with the goal of generating as many diverse ideas as possible. There will be time to think critically, analyze, and filter ideas later. For now, try to generate as many ideas as possible. Most of them may end up being unoriginal or just not very good – but if you come up with enough ideas, there's sure to be a gem in there somewhere!



“The best way to have a good idea is to have a lot of ideas.”

-Linus Pauling

All About Brainstorming

There are many tools for successful ideation, but one of the most frequently-used and effective methods is **brainstorming**: a semi-structured, team-based approach to rapid idea generation.

Each term in this definition is important:

- **Semi-structured**

Brainstorming isn't a free-for-all! It has a structure and specific rules for everyone to follow. Targeted creativity needs some constraints. (This also is why we frown upon need statements like "A way to cure cancer..." – they're too broad, without enough specificity to lead to tangible solutions.) At the same time, a good brainstorm is improvised and has no pre-defined path. Brainstorming resides in the middle ground between anarchy and a well-planned meeting.

- **Team-based**

Effective brainstorming is a "team sport." By design, it engages participants with a wide range of backgrounds and points of view (more on this below). It creates an environment where they can build off one another's ideas, tapping into the creative potential of the entire group and resulting in a whole that is greater than the sum of its parts.

- **Rapid idea generation**

As noted, brainstorming isn't about going deep on a few ideas, it's about generating a vast quantity of ideas in order to find the best one(s). The key is to expand the realm of what's possible rather than anchoring on the first or most obvious ideas you come up with.

In practice, brainstorming is conducted as a group meeting where participants share ideas on a focused problem area. The ideas are placed on a shared space, usually a whiteboard or wall, so that the accumulation of ideas can be visualized – which in itself can spark new ideas. If you've never participated one, check out the video called Sample Brainstorming Session to get a general sense for how a brainstorm works.

While this is the core of the brainstorm, there's a lot that goes on before, during, and after the session(s) that can increase your chances of success. And there are a number of tried-and-true brainstorming rules that can ensure the session runs smoothly and maximize the creative potential of the team. Read on!

Brainstorming Rules

Brainstorming sessions work best if all of the participants understand and follow the same set of rules. These seven rules are intended to optimize the creative output that results from the session:¹

- **#1 – Defer judgement**

Don't dismiss any ideas. The goal is to be generative, not analytical. If you find yourself thinking an idea is implausible, one way to prevent yourself from shooting it down is, instead, to think about what you would need to do to make the idea work (leveraging rules 2 and 3 below).

- **#2 – Encourage wild ideas**

Great ideas often hide right behind the crazy, unrealistic ones. And sometimes a crazy idea triggers someone else to come up with a great idea. So don't hold back!

- **#3 – Build on the ideas of others**

Use the words: "Yes, AND...!" to help you expand and enhance the ideas of others. It's often helpful to spend a bit of time on each idea, building off it as much as possible,

before proceeding to the next idea or theme. This is one of the most important rules, as it's the whole reason you get together as a group to brainstorm!

- **#4 – Go for quantity**

Try for one idea per minute! This will keep your session moving at a good pace.

- **#5 – One conversation at a time**

Listening is as important as talking. And it's essential to being able to build off each other's ideas.

- **#6 – Stay focused on the topic**

There's a fine line between getting creative and going off topic. Stay true to the intended focus of each brainstorming session by designating a "Parking Lot." This is where you can capture interesting ideas or new lines of thought that aren't directly related to the current topic. Then you can revisit this information after the session.

- **#7 – Be visual**

In a classic brainstorming session, everyone should be holding a pen and ready to capture and contribute their ideas. Ideally, the capture will include a short headline and a quick doodle to help you communicate and remember the idea. Don't worry about your drawing skills. Sometimes even a misinterpretation of what you intended can trigger great ideas in others!

Check out the videos called Demo of an Effective Brainstorm and Demo of an Ineffective Brainstorm for examples of how a team can generate some great ideas when they follow these rules and what happens when they don't!

Participant Selection

Ideally, your team already includes a mix of engineers and scientists from different specialties. It may also include medical providers and/or folks with business education/experience. If not, try to identify and invite people with points of view your team is lacking to join you for one or more brainstorming sessions. In addition, consider including one or two people with early-stage, low-fidelity prototyping experience. This can be beneficial when it comes to rapidly conceiving and bringing new ideas to light.

If you're considering inviting outsiders to your brainstorm session, think about diversity in terms of education, professional expertise, socioeconomics, demographics, etc. Assembling a well-rounded group allows for the cross-pollination of ideas from different fields and perspectives, which is a common recipe for successful innovation. Ideas that are considered conventional in one field could be radically transformative in another (e.g., the use of balloons in arteries or the use of lasers in ophthalmology).

Counterintuitively, involving a senior expert from the specific field you're working in can backfire. They might be reflexively conservative and critical of new ideas, intentionally or unintentionally protecting the status quo. They may also be more focused on incremental ideas that they can immediately visualize, and less supportive of big, transformative ideas that may initially seem outlandish. Less senior team members may be tempted to defer to the opinions of the expert, rather than contributing as equals. It's often more valuable to access senior experts later in the process, when the team's concepts have been developed in more detail, so that senior experts can more objectively and thoroughly evaluate them.

If your team decides to invite outside participants to a brainstorming session, you'll want to set clear expectations in advance about any IP that's potentially generated. For example, if the invitee is an employee of a company that's in the same field (or even an adjacent field), it could create a conflict if they help conceive of an idea that the team decides to take forward. Usually, it's a good idea to ask outside participants to sign confidentiality agreements. And they may be willing to formally waive their rights to any inventions that come out of the session. Check with your university's Office of Technology Transfer for guidance and/or standard forms that you may be able to leverage.

Number of Participants

The ideal number of participants in a brainstorming session can vary, but 4-6 is a good target. With fewer than four people, you risk not having enough stimulating input to build off the ideas of each other. With seven or more people, brainstorming can become unwieldy and difficult for each individual to make a substantive contribution.

Duration

When it comes to the length of a brainstorming session, shorter is usually better. Brainstorming can be tiring, and participants will find their creativity and energy drained if they try to sustain high engagement levels for too long. It's often more productive to do three 30-minute brainstorming sessions versus one 90-minute session. Many teams find 45 minutes to be optimal (see Box 1 for a sample schedule).

BOX 1 A Typical 45-Minute Brainstorming Schedule

Here's an example of how you might spend your time in a 45 minute session:

0:00-0:05 Expectation setting and warm-up

A warm-up gets everyone's creative juices flowing! Set aside 3- 5 minutes for an activity that puts participants at ease and gets them into an open mindset. You also should lay-down ground rules and get everyone on the same page about the goals of the session.

0:05-0:35 Brainstorming

This is where the magic happens. Careful preparation will help you ensure you make the most of your brainstorming time.

0:35-0:45 Idea capture and discussion

Be sure to allocate time to the careful review and capture of your ideas at the end of the session. Before breaking, it's helpful to have participants reflect on the ideas that the group generated and discuss where they still think gaps may exist.

You can find more information below about warm-ups, planning and conducting a brainstorm, and what happens after the session.

Materials/Supplies

Having the right tools for any job always makes things easier. Below is a short list of materials that can enhance the productivity of your brainstorms:

- **Copious writing space**

The preferred modern medium for brainstorming is the whiteboard, though even a blank wall could suffice in a pinch (if using sticky notes). The key is that you have enough space so that you can visualize all of the ideas generated in a session simultaneously without having to erase or remove any of them. This also is helpful as you get toward the end of a session so you can group common ideas together, make sure there are no repeats or major gaps, and categorize ideas by theme or approach.

- **Post-It® notes**

These little pieces of colorful, sticky paper have become practically synonymous with brainstorming. There are a variety of benefits of using sticky notes in your brainstorms: (1) You can move ideas around the board, recategorizing them as new ideas are generated; (2) You can color-code ideas by person or theme, (3) When "solo storming" (see below), you can write your notes down on sticky notes and apply them directly during the brainstorm to save time; and (4) they require you to be concise.



Photo by Stanford Biodesign

- **Toys / prototyping materials / anatomical models**

Physical artifacts are useful for many reasons. Toys and rapid prototyping materials (like clay, LEGO® blocks, pipe cleaner, other arts and craft materials, etc.) can trigger ideas and provide a tactile mode of thinking and visualizing ideas. If brainstorming around a specific anatomical region, models can be helpful to visualize the 3-dimensional space. Similarly, if targeting a specific disease process, a diagram of the pathophysiology can be instructive and trigger ideas of where and how to intervene.

- **Candy / snacks / water / caffeine**

Every brainstorm needs fuel to keep folks energized! Having drinks and easy-to-eat snacks on hand can keep you powering through the brainstorm session.

Planning a Brainstorm: Before, During, and After

Now that you understand the fundamentals of brainstorming, it's time to think about how to plan and hold a session – or, better yet, a series of sessions. For most projects, it's best not to think about brainstorming as a one-time event, but instead something you'll do multiple times, with different jumping off points, in an effort to generate the broadest possible range of diverse ideas.

What do we mean by a "jumping off point"? Each session should have one or more specific prompts to stimulate ideation. One effective method, borrowed from design thinking, is to develop How Might We (HMW) questions.² Take specific insights you have from your need research, and reframe them into questions starting with "How might we..." For example, the team working on a

way to address night terrors in children might have asked, “How might we disrupt night terrors using electrical energy?” Here are some other ways to think about coming up with HMW questions:

- **Different mechanisms of action or root causes**

Leverage what you’ve learned about the pathophysiology of the relevant condition to generate HMW questions. For instance, with the night terrors example, you could ask how might we positively affect the non-REM sleep cycle to avoid or interrupt night terrors? Or HMW cause the child to bypass the night terror in the transition from non-REM to REM sleep? Or HMW affect brain chemicals during sleep to prevent night terrors? If you’re working on a need that’s more of a health system issue, look at a root-cause analysis diagram for inspiration.

- **Different forms of energy**

Another way to think about HMW questions is to consider the different forms of energy you could use to have a desired effect. For example, HMW disrupt night terrors using mechanical energy? Or electrical, thermal, chemical, etc.

- **Cycle of care**

You can also think about intervening at different points in the cycle of care (see the toolkit called Scoping and Finalizing Your Need Statement for more information on cycle of care). This is particularly useful for healthcare problems that involve a complicated care or referral path, or involve a large number of different stakeholders. For the night terror example, you can brainstorm solutions early in the diagnosis process with questions like HMW empower a parent to differentiate between night terrors and nightmares? Or HMW raise awareness of night terrors among parents whose children have sleeping issues? To ideate further down the care continuum, you might try questions such as HMW treat night terrors in a way that doesn’t involve the parents (so they can get a good night’s sleep)?

- **Different venues**

Changing the venue is another good way to trigger ideas, especially when taking solutions from more intensive settings of care (e.g., ICU, operating room, emergency room) to less intensive settings (e.g., doctor’s office, nursing home, patient’s home). This approach is less relevant with a home-based need like night terrors, but highly relevant when dealing with a surgical or procedure-based need.

- **Other constraints**

Generally speaking, applying different constraints can have the effect of triggering widely divergent ideas. For example, HMW accomplish a task without any electricity, or automatically (without human intervention), or remotely? By enforcing constraints, it forces your brain to come up with more innovative solutions that might just reveal the next great idea.

When developing HMW question you want to strike a balance – the questions should be narrow enough to give you a clear starting point, but broad enough that you have room to explore a wide range of ideas.³ An example of a HMW question that’s too abstract might be, “HMW help people with diabetes?” Similar to a need statement that’s too broad, it lacks enough constraints to inspire creativity. On the other extreme, “HMW make the buttons on a continuous glucose monitor bigger for diabetics with impaired mobility?” is an example of a HMW question that’s too narrow. Overly focused HMW statements are too prescriptive and tend to have solutions embedded in them, which limits the creative potential of the group.

If needed, you can spend the first ~10 minutes of your brainstorming session coming up with HMW statements. But, more often than not, it can work even better to hold a separate HMW brainstorm where you and your team are solely focused on brainstorming and then prioritizing different HMW questions. Watch the video called Coming Up with “How Might We...” Questions for an example. Cluster the best HMWs into a set of prompts that you think would work well for each session (usually 2-3 prompts per session) and then decide how many sessions you need to hold to cover the best ones. It’s not unusual for a team to hold 3-5 brainstorming sessions for a class project.

Other Activities Before the Brainstorm

Once you have a vision for the brainstorming sessions you’ll hold, designate one team member as the **facilitator** for each session. The facilitator helps ensure a smooth brainstorm meeting by doing the planning and prep work, running the session, and enforcing the rules of brainstorming.

The facilitator’s first job is to take care of **session logistics**. The key activities here are to choose a date and time, decide on a duration that’s appropriate for addressing the defined prompts, and create a detailed agenda for how you’ll use the time. Additionally, find a location that’s conducive to brainstorming, identify and select diverse participants, and get the sessions on everyone’s calendars.

Next, the facilitators should make sure everyone is prepared for the session since preparation is paramount for brainstorming. The participants all should have a good understanding of the background research that formed the basis for the need statement and need criteria. For any invited participants not directly involved with the project, send a summary of your need research ahead of time as **pre-reading material** to enable everyone to hit the ground running during the brainstorm itself.



“Chance favors the prepared mind.”

-Louis Pasteur

In addition, studies have shown that having participants conduct a **solo storm** on their own, prior to the brainstorming session, can significantly enhance productivity and both the quantity and quality of ideas. There are a number of ways to solo storm. The most straightforward method is to ask each participant to come up with 10 ideas for each of your prompts or HMW questions before the meeting – preferably with each idea captured on an individual sticky note. Then, in the session, get started by having each person briefly present their ideas for the first prompt – putting their sticky notes on the wall, and discussing each one briefly. As this happens, allow others to add similar or related ideas to create synergy. It’s far less productive if everyone just dumps their ideas all at once and puts up their sticky notes in rapid succession without building off the ideas of others.

The facilitator should also decide on a **warm-up exercise** to start the session. Warm-up helps everyone become more relaxed and gets the creative juices flowing. Warm-ups can be physical in

nature (getting up, stretching, jumping jacks, etc.), mental, or a combination of both. Many ideas are available online, but here are a few examples:

- **Rock-paper-scissors tournament**

This approach can be especially fun with a larger group (for example, if you have multiple groups warming up together before splitting off into different brainstorming sessions). This exercise is a perfect metaphor for brainstorming as it requires a flexible mind – you're competitive at first, but when you lose a round you then have to flip and be the #1 fan of the person you just lost to, which encourages a collaborative mindset.

- **Alternate uses**

This warm-up is based on a test used in psychology experiments to measure people's creativity. Select a common household object (like a paperclip or toothbrush) and individually try to come up with as many alternate uses for it as possible. The number of diverse categories of use is one indicator of creativity. Then share with the group.

- **30 circles**

For this exercise, give everyone a piece of paper with 30 blank circles on it. In three minutes, using nothing but a pen, ask them to turn as many of the circles as possible into recognizable objects (e.g., happy face, car tire). This gets everyone thinking visually, thinking outside the box, and comfortable with drawing.

Finally, it's a good idea for the facilitator to be intentional about how to **set up and use the brainstorming space**. Put your need statement in big letters at the top of your writing space, so you can refer to it throughout the brainstorm. It can also be helpful to post the seven brainstorming rules for reference during the session. Think about how many prompts you plan to tackle and whether you have enough room to set up a different writing space for each one. If not, you'll need to be able to capture and then move your notes after each prompt – in which case, you might want to put your sticky notes on flip chart paper so they can quickly be photographed and then moved to the side to make space without losing any context around how the notes were positioned and placed.

During the Brainstorm

The main role of the facilitator during the session is to guide the group through the agenda while upholding the rules of brainstorming. The trick is to allow for unexpected and creative twists and turns but also prevent the group from going too far afield or lingering on one idea for too long to ensure forward momentum.

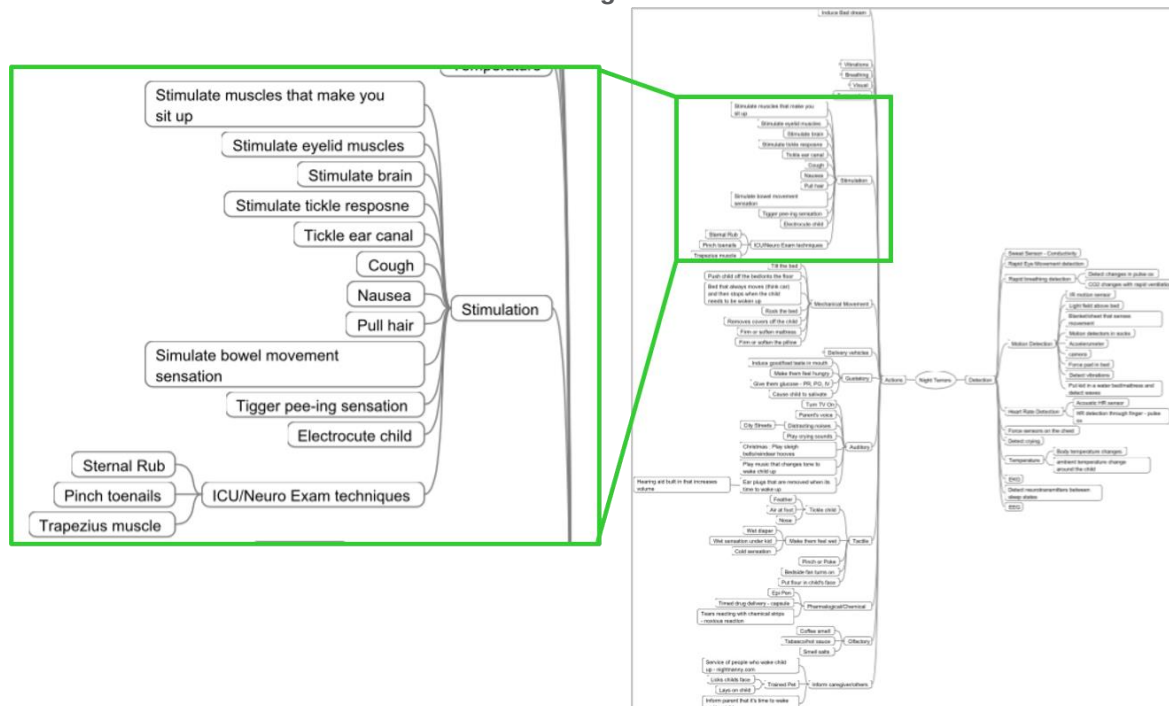
If you have outside participants, the facilitator can sometimes feel a bit uncomfortable enforcing the rules – especially if they're more senior than you in terms of position or experience. One diplomatic way to do this is explicitly tell everyone that's your role, review the rules before getting started, and then refer back to them if you have to intervene. Try saying things like "Let's remember the rules and try not to criticize ideas. How can we build on them instead?" or "I love where you're going with that idea, but it's a bit off topic. I'll make a note in our Parking Lot so we don't forget to come back to it later."

Doing a good job of facilitation takes plenty of focus and energy. That's why it generally makes sense for that person not to be an active participant in the brainstorm. They can bring a few ideas into the session to help stimulate ideas, but then should let the group do the ideation while they stay focused on the flow, energy, and output of the session.

After the Brainstorm

As the session is winding down, the facilitator should take the lead in capturing the output of the session. One good way to do this is engage the group in organizing the ideas on the board by thematic area or approach in a mind map (e.g., energy modality, technology categories, anatomical approach, mechanism of action). This is easiest if you have sticky notes that you can move around (but you can also do this later using mind mapping software, e.g., [Miro](#) or [Lucidchart](#)). The benefit of doing this together, before the session is over, is that it gives everyone the opportunity to reflect. In particular, look for gaps in the ideas that the group generated. Did you miss a potentially important “zone” for new solutions? If so, reconnect with the team to decide how to address the gap in another upcoming session. A portion of a mind map from the night terrors team is shown in Figure 1.

Figure 1 – A Sample Mindmap;
The Callout Shows Ideas to Address the Question HMW Use Stimulation to Interrupt or Prevent a Night Terror?



Note: The callout shows ideas to address the question “HMW Use Stimulation to Interrupt or Prevent a Night Terror?” (courtesy of the Lully team)

You should also make sure you can remember what each idea is about. Will the short headlines on the sticky notes make sense after everyone leaves the room? If not, ask the group to help you clarify the intended meaning by jotting down a few additional notes or the name of the person who generated the idea on the back of the Post-it so you can follow up.

Finally, take plenty of pictures, zooming in as needed. You don’t want to miss a promising idea because you tried to capture the whole board in one photo that wasn’t high-resolution enough!

Watch the video called Brainstorming Tips for other pointers related to what happens before, during, and after a brainstorm.

Next Up

Teams have two main works streams to pursue after brainstorming. One is to move into concept screening, where you start to assess and prioritize the ideas you've generated – first by comparing them against the need criteria and then investigating them relative to other factors such as intellectual property, regulation, payment/reimbursement, and business model (see the Concept Screening and Selection toolkit). Another is to assess the technical feasibility of your ideas, which requires you to consider the ideas at a comparable level of detail (see the After Brainstorming...Now What? toolkit).

In the meantime, have fun with brainstorming and enjoy the creative process!

Credits

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Notes

¹ Tom Kelley, *The Art of Innovation*, Currency, 2001.

² How Might We Design Kit, IDEO.org, <https://www.designkit.org/methods/3> (May 27, 2021).

³ Ibid.